# Pizza Sales — SQL Analysis Report

Database: pizzasales | Tool: MySQL | Dataset: 2015

*Prepared by: Ashish Sharma | Project #3 — Data Analysis*

This report documents all SQL queries written for the Pizza Sales Analysis project, along with the actual results obtained by running each query against the dataset. The analysis covers 21,350 orders, 48,620 order detail records, 32 pizza types, and 96 pizza variants across the year 2015.

## Q1. Retrieve the total number of orders placed

### SQL Query

```
SELECT COUNT(order_id) AS total_orders
FROM orders;
```

### Result

| Total Orders Placed | 21,350 |
|---|---|

> *The pizza shop received 21,350 orders throughout 2015 — averaging roughly 58 orders per day across the year.*

## Q2. Calculate the total revenue generated from pizza sales

### SQL Query

```
SELECT ROUND(SUM(orders_details.quantity * pizzas.price), 2)
       AS total_sales
FROM orders_details
JOIN pizzas ON pizzas.pizza_id = orders_details.pizza_id;
```

### Result

| Total Revenue Generated | $817,860.05 |
|---|---|

> *The shop generated $817,860.05 in total sales in 2015, averaging approximately $2,241 in revenue per day.*

## Q3. Identify the highest-priced pizza

### SQL Query

```
SELECT pizza_types.name, pizzas.price
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

### Result

| Pizza Name | Price |
|---|---|
| The Greek Pizza | $35.95 |

> *The Greek Pizza is the most premium item on the menu at $35.95, which is significantly higher than the average pizza price.*

## Q4. Identify the most common pizza size ordered

### SQL Query

```
SELECT pizzas.size,
       COUNT(orders_details.order_details_id) AS order_count
FROM pizzas
JOIN orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

### Result

| Pizza Size | Order Count |
|---|---|
| L (Large) | 18,526 |
| M (Medium) | 15,385 |
| S (Small) | 14,137 |
| XL (Extra Large) | 544 |
| XXL (Double XL) | 28 |

> *Large (L) is by far the most popular size with 18,526 orders — nearly 39% of all orders. XXL is rarely ordered with just 28 orders all year.*

## Q5. List the top 5 most ordered pizza types along with their quantities

### SQL Query

```
SELECT pizza_types.name,
       SUM(orders_details.quantity) AS quantity
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

### Result

| Rank | Pizza Name | Total Quantity Ordered |
|---|---|---|
| 🥇 1 | The Classic Deluxe Pizza | 2,453 |
| 🥈 2 | The Barbecue Chicken Pizza | 2,432 |
| 🥉 3 | The Hawaiian Pizza | 2,422 |
| 4 | The Pepperoni Pizza | 2,418 |
| 5 | The Thai Chicken Pizza | 2,371 |

> 💡 *The Classic Deluxe Pizza leads in quantity with 2,453 units sold. The top 5 pizzas are surprisingly close in numbers — only 82 units separate rank 1 from rank 5.*

## Q6. Total quantity of each pizza category ordered

### SQL Query

```
SELECT pizza_types.category,
       SUM(orders_details.quantity) AS quantity
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

### Result

| Category | Total Quantity Ordered |
|---|---|
| Classic | 14,888 |
| Supreme | 11,987 |
| Veggie | 11,649 |
| Chicken | 11,050 |

> 💡 *Classic pizzas dominate with 14,888 units — about 30% of all orders. All four categories are fairly balanced, suggesting the menu has something for every type of customer.*

## Q7. Distribution of orders by hour of the day

### SQL Query

```
SELECT HOUR(order_time) AS hour,
       COUNT(order_id) AS order_count
FROM orders
GROUP BY HOUR(order_time);
```

### Result

| Hour | Order Count | Observation |
|---|---|---|
| 9 AM | 1 | Shop opens |
| 10 AM | 8 | Very slow |
| 11 AM | 1,231 | Lunch rush begins |
| 12 PM | 2,520 | 🔥 Peak hour |
| 1 PM | 2,455 | 🔥 Peak hour |
| 2 PM | 1,472 | Post-lunch drop |
| 3 PM | 1,468 | Afternoon slow |
| 4 PM | 1,920 | Picks up |
| 5 PM | 2,336 | Dinner rush begins |
| 6 PM | 2,399 | 🔥 Peak hour |
| 7 PM | 2,009 | Dinner tapering |
| 8 PM | 1,642 | Evening wind-down |
| 9 PM | 1,198 | Late evening |
| 10 PM | 663 | Closing time |
| 11 PM | 28 | Last orders |

> 💡 *There are two clear peak periods — Lunch (12 PM–1 PM) and Dinner (5 PM–6 PM). 12 PM is the single busiest hour with 2,520 orders. The shop is virtually empty before 11 AM.*

## Q8. Category-wise distribution of pizza types

### SQL Query

```
SELECT category, COUNT(name) AS pizza_count
FROM pizza_types
GROUP BY category;
```

### Result

| Category | Number of Pizza Types |
|---|---|
| Chicken | 6 |

| Category | Number of Pizza Types |
|---|---|
| Classic | 8 |
| Supreme | 9 |
| Veggie | 9 |

> 💡 *Supreme and Veggie categories each have the most variety (9 types), while Chicken has the least options (6 types) yet still sells 11,050 units — showing strong demand per type.*

## Q9. Average number of pizzas ordered per day

### SQL Query

```
SELECT ROUND(AVG(quantity), 0) AS avg_pizza_ordered_per_day
FROM (
  SELECT orders.order_date,
         SUM(orders_details.quantity) AS quantity
  FROM orders
  JOIN orders_details ON orders.order_id = orders_details.order_id
  GROUP BY orders.order_date
) AS order_quantity;
```

### Result

| Average Pizzas Ordered Per Day | 138 pizzas/day |
|---|---|

> 💡 *On average, 138 pizzas were ordered every single day in 2015. This is a useful baseline for inventory planning and daily prep estimates.*

## Q10. Top 3 most ordered pizza types based on revenue

### SQL Query

```
SELECT pizza_types.name,
       SUM(orders_details.quantity * pizzas.price) AS revenue
FROM pizza_types
JOIN pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

### Result

| Rank | Pizza Name | Revenue |
|---|---|---|
| 🥇 1 | The Thai Chicken Pizza | $43,434.25 |
| 🥈 2 | The Barbecue Chicken Pizza | $42,768.00 |

| Rank | Pizza Name | Revenue |
|---|---|---|
| 🥉 3 | The California Chicken Pizza | $41,409.50 |

> 💡 *Interesting! Chicken pizzas dominate the top 3 by revenue even though Classic leads in quantity. This means Chicken pizzas are priced higher — they generate more money per order.*

## Q11. Percentage contribution of each pizza category to total revenue

### 📝 SQL Query

```
SELECT pizza_types.category,
       ROUND(SUM(orders_details.quantity * pizzas.price) /
             (SELECT SUM(orders_details.quantity * pizzas.price)
              FROM orders_details
              JOIN pizzas ON pizzas.pizza_id = orders_details.pizza_id)
             * 100, 2) AS revenue_percentage
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue_percentage DESC;
```

### 📊 Result

| Category | Revenue Contribution (%) |
|---|---|
| Classic | 26.91% |
| Supreme | 25.46% |
| Chicken | 23.96% |
| Veggie | 23.68% |

> 💡 *Revenue is remarkably balanced across all four categories — no single category dominates. Classic leads with 26.91% but Veggie is just 3.23 percentage points behind, showing a well-diversified menu.*

## Q12. Cumulative revenue generated over time

### 📝 SQL Query

```
SELECT order_date,
       ROUND(SUM(revenue) OVER(ORDER BY order_date), 2)
       AS cumulative_revenue
FROM (
  SELECT orders.order_date,
         SUM(orders_details.quantity * pizzas.price) AS revenue
  FROM orders_details
  JOIN pizzas ON orders_details.pizza_id = pizzas.pizza_id
  JOIN orders ON orders.order_id = orders_details.order_id
```

```
  GROUP BY orders.order_date
) AS sales;
```

### 📊 Result (First 10 days shown — full query returns all 365 days)

| Date | Cumulative Revenue ($) |
|---|---|
| 2015-01-01 | $2,713.85 |
| 2015-01-02 | $5,445.75 |
| 2015-01-03 | $8,108.15 |
| 2015-01-04 | $9,863.60 |
| 2015-01-05 | $11,929.55 |
| 2015-01-06 | $14,358.50 |
| 2015-01-07 | $16,560.70 |
| 2015-01-08 | $19,399.05 |
| 2015-01-09 | $21,526.40 |
| 2015-01-10 | $23,990.35 |

> 💡 *This query uses a Window Function (SUM OVER ORDER BY) to calculate running totals. By end of Day 10, the shop had already crossed $23,990. The full query tracks growth all the way to $817,860 by year end.*

## Q13. Top 3 most ordered pizza types by revenue for each category

### 📝 SQL Query

```
SELECT name, revenue FROM (
  SELECT category, name, revenue,
         RANK() OVER(PARTITION BY category ORDER BY revenue DESC) AS rn
  FROM (
    SELECT pizza_types.category, pizza_types.name,
           SUM(orders_details.quantity * pizzas.price) AS revenue
    FROM pizza_types
    JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN orders_details ON orders_details.pizza_id = pizzas.pizza_id
    GROUP BY pizza_types.category, pizza_types.name
  ) AS a
) AS b
WHERE rn <= 3;
```

### 📊 Result

| Category | Pizza Name | Revenue |
|---|---|---|
| 🍗 Chicken | The Thai Chicken Pizza | $43,434.25 |
| 🍗 Chicken | The Barbecue Chicken Pizza | $42,768.00 |
| 🍗 Chicken | The California Chicken Pizza | $41,409.50 |
| 🍕 Classic | The Classic Deluxe Pizza | $38,180.50 |

| Category | Pizza Name | Revenue |
|---|---|---|
| 🍕 Classic | The Hawaiian Pizza | $32,273.25 |
| 🍕 Classic | The Pepperoni Pizza | $30,161.75 |
| 👑 Supreme | The Spicy Italian Pizza | $34,831.25 |
| 👑 Supreme | The Italian Supreme Pizza | $33,476.75 |
| 👑 Supreme | The Sicilian Pizza | $30,940.50 |
| 🌿 Veggie | The Four Cheese Pizza | $32,265.70 |
| 🌿 Veggie | The Mexicana Pizza | $26,780.75 |
| 🌿 Veggie | The Five Cheese Pizza | $26,066.50 |

> 💡 *This query uses RANK() with PARTITION BY — an advanced Window Function. It ranks pizzas within each category independently. Thai Chicken ($43,434) is the single highest-revenue pizza across all categories.*

# 📋 Project Summary

| Metric | Value |
|---|---|
| Total Orders | 21,350 |
| Total Revenue | $817,860.05 |
| Average Revenue Per Day | ~$2,241 |
| Average Pizzas Per Day | 138 |
| Most Ordered Pizza | The Classic Deluxe Pizza (2,453 units) |
| Highest Revenue Pizza | The Thai Chicken Pizza ($43,434.25) |
| Most Popular Size | Large — L (18,526 orders) |
| Highest Priced Pizza | The Greek Pizza ($35.95) |
| Peak Order Hour | 12 PM (2,520 orders) |
| Top Revenue Category | Classic (26.91% of total revenue) |
| SQL Concepts Used | JOINs, GROUP BY, Subqueries, Window Functions (RANK, SUM OVER) |

*This project involved writing 13 SQL queries of varying complexity — from basic aggregations to advanced window functions using RANK() and SUM() OVER(). All queries were executed on a MySQL database (pizzasales) loaded with real 2015 pizza sales data.*

**— Ashish Sharma | Aspiring Data Analyst | Project #3**